

WENRUI XU

Minneapolis, Minnesota

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Education

University of Minnesota, Twin Cities(Advisor: Prof. Keshab. K. Parhi)

Expect 2028

Doctor of Philosophy in Electrical Engineering (GPA: 3.9/4.0)

Minneapolis, Minnesota

University of Minnesota, Twin Cities

May 2023

Bachelor of Computer Engineering(GPA: 3.9/4.0)

Minneapolis, Minnesota

Work Experience

Boston Scientific

May 2025 – August 2025

Rhythm Management R&D Research Intern

Arden Hill, Minnesota

- Developed recurrent neural network (RNN)-based models to predict clinical outcomes of pacemaker patients, leveraging multi-year ECG and heart rate data.
- Designed and optimized sequential models that improved predictive accuracy by capturing long-term physiological patterns.

Boston Scientific

May 2024 – August 2024

Rhythm Management R&D Research Intern

Arden Hill, Minnesota

- Researched and prototyped novel device innovations for the cardiac rhythm management and diagnostics team.
- Built a Generative AI API-integrated GUI tool that automated academic literature summarization, streamlining research workflows.

Boston Scientific

May 2023 – August 2023

Rhythm Management R&D Research Intern

Arden Hill, Minnesota

- Designed convolutional neural networks (CNNs) to classify arrhythmias from ECG signals with improved diagnostic accuracy.
- Engineered a parallelized preprocessing algorithm that reduced computation time for a 2.5 TB ECG database by 40×.

Research Experiences

Graduate Research Assistant, with Prof. Keshab K. Parhi

September 2023 - September 2024

• Knowledge Graph Project:

- Apply the Hyperdimensional Computing (HD) method to encode knowledge graphs using FB15K-237 and WN18RR database.
- Prove the applicability of the HD method on knowledge graph completion tasks.

Undergraduate Research Assistant, with Prof. Keshab K. Parhi

June 2022 - May 2023

• Arrhythmia Detection Project

- Apply the Hyperdimensional Computing (HD) method to classify arrhythmia using the MIT-BIH database.
- Manually extract FFT (Fast Fourier Transform), HRV, and DWT (Discrete Wavelet Transform) features, and apply feature selection algorithms to identify the top N features for the HD model training.
- Validate the applicability of the HD method for detecting multiple abnormal heartbeats types.

• Seizure Detection Project:

- Apply the features of LBP (Local Binary Pattern) and PSD (Power Spectral Density) from the CHB-MIT Database.
- Train and cross-validate HD models on ictal phase data of different seizure types and optimize HD models for seizure prediction.
- Prove the applicability of the HD method on UPenn and Mayo Clinic's Seizure Detection Challenge.

Technical Skills

Programming: Python, C, Java, JavaScript, Assembly, Matlab, SQL, HTML/CSS

ML/AI: TensorFlow, PyTorch, Hyperdimensional Computing

Tools: Git, Conda

Activities

Engineering Honor Society, Tau Beta Pi

Spring 2022 — Present

IEEE Student Branch

Spring 2022 — Present

HKN Omicron Chapter

Spring 2022 — Present

Solar Vehicle Project

Summer 2021 — Summer 2023